$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$\frac{\partial L}{\partial z} = \frac{\partial L}{\partial \sigma(z)} \times \frac{\partial(\sigma(z))}{\partial z}$$

$$L = -\left[y \log(\sigma(z)) + (1-y)\log(1-\sigma(z))\right]$$

$$\frac{\partial L}{\partial \sigma(z)} = -\left[\frac{y}{\sigma(z)} + \frac{1-y}{1-\sigma(z)}\right]$$

$$= \frac{\sigma(z)-y}{\sigma(z)(1-\sigma(z))}$$

$$\frac{\partial \sigma(z)}{\partial z} = \sigma(z)(1-\sigma(z))$$

$$\frac{\partial L}{\partial z} = \frac{\sigma(z)-y}{\cancel{\sigma(z)}(\cancel{1-\sigma(z)})} \times \cancel{\sigma(z)}(\cancel{1-\sigma(z)})$$

$$= \sigma(z) - y$$

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial z} \times \frac{\partial z}{\partial w}$$

$$= \sigma(z) - y \times x$$

$$-\frac{\partial L}{\partial w} = (y - \sigma(z))x$$

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial z} \times \frac{\partial z}{\partial w}$$

$$= \sigma(z) - y$$

$$-\frac{\partial L}{\partial w} = (y - \sigma(z))$$

Multi-class logistic regression:

$$B = 2 \quad \text{(Batch Size)}$$
$$N = 3 \quad \text{(Features)}$$
$$C = 3 \quad \text{(Classes)}$$

Input Matrix X (2×3):

$$X = \underbrace{\begin{bmatrix} x_{1,1} & x_{1,2} & x_{1,3} \\ x_{2,1} & x_{2,2} & x_{2,3} \end{bmatrix}}_{B \times N} \qquad \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}$$

Weight Matrix W (3×3):

$$W = \underbrace{\begin{bmatrix} w_{1,1} & w_{1,2} & w_{1,3} \\ w_{2,1} & w_{2,2} & w_{2,3} \\ w_{3,1} & w_{3,2} & w_{3,3} \end{bmatrix}}_{N \times C} = \begin{bmatrix} W_1 & W_2 & W_3 \end{bmatrix}$$

(Note: Columns correspond to classes 1, 2, and 3) Bias Vector b (1×3):

$$b = \begin{bmatrix} b_1 & b_2 & b_3 \end{bmatrix}_C$$

Expanded Equation $Z = XW + b$:

$$Z = \begin{bmatrix} (x_{1,1}w_{1,1} + x_{1,2}w_{2,1} + x_{1,3}w_{3,1}) + b_1 & (x_{1,1}w_{1,2} + x_{1,2}w_{2,2} + x_{1,3}w_{3,2}) + b_2 & (x_{1,1}w_{1,3} + x_{1,2}w_{2,3} + x_{1,3}w_{3,3}) + b_3 \\ (x_{2,1}w_{1,1} + x_{2,2}w_{2,1} + x_{2,3}w_{3,1}) + b_1 & (x_{2,1}w_{1,2} + x_{2,2}w_{2,2} + x_{2,3}w_{3,2}) + b_2 & (x_{2,1}w_{1,3} + x_{2,2}w_{2,3} + x_{2,3}w_{3,3}) + b_3 \end{bmatrix}$$

$$= W_1 X_1 + b_1 \qquad W_2 X_2 + b_2 \qquad W_3 X_3 + b_3$$

This is how we end up with multiple decision boundaries